// PROCUREMENT INTELLIGENCE CASE STUDY

Meet Aldermoor Industries

The company, the purchase, and the people who must say yes

Aldermoor Industries, its people and its documents are fictional.



// 01 THE COMPANY

A company that buys before it builds

A machine builder

Builds machines that fill and pack bottles and boxes.

Based in Hamburg

Headquartered in Hamburg, Germany, with more than 40 subsidiaries worldwide.

Buys from the world

Servo motors, controllers, sensors, steel, food-grade chemicals and packaging materials.

Fictional by design

Every person, figure and document in this project is made up for learning.

> The pattern is real: a company with many suppliers, strict internal rules and documents that hold the answers.

// 01 THE COMPANY

A typical purchase request

purchase-request.txt

VENDOR	Takumi Controls Europe B.V. (Netherlands)
ITEM	Industrial PLC controllers and HMI panels, quantity 200
TOTAL COST	EUR 480,000
ORIGIN	Japan
DESTINATION	Aldermoor Industries, Hamburg, Germany
QUOTED RATE	1 EUR = 163.5 JPY

// WHAT MAKES IT DEMANDING

A foreign supplier
Is it allowed to trade with? Can it be trusted?

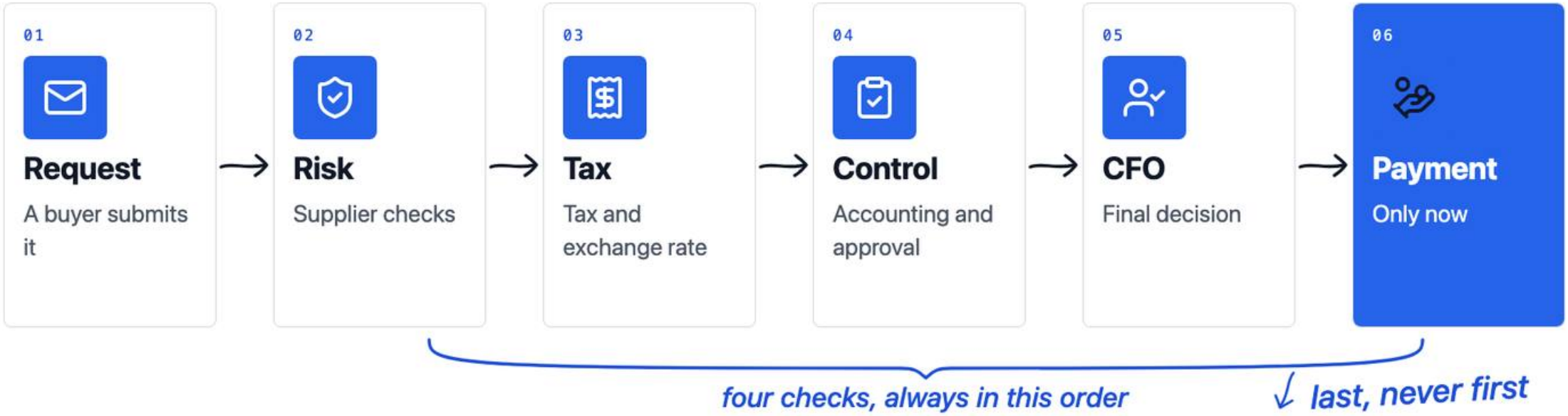
Crossing a border
Import VAT and customs duty apply.

A large amount
Someone senior has to approve it.

Another currency
Is the quoted exchange rate fair?

// 01 THE COMPANY

Nothing is paid until it is checked



// 01 THE COMPANY

Four departments, four questions

Risk and compliance
Is this supplier allowed? Can it be trusted to deliver and to survive financially?

Tax and treasury
How much tax and import duty will we pay? Is the exchange rate in the quote sensible?

Finance and control
Is this a lasting asset or a running cost? Who has the authority to approve this amount?

The CFO
Taking everything together: do we pay, hold or reject?

What checked means in practice



Risk and compliance

- Screen the supplier against sanctions lists
- Credit score 75 or more: low risk
- 50 to 74: deposit or letter of credit
- Below 50: escalate to the CFO



Tax and treasury

- Import VAT of 19 percent in Germany
- Customs duty by tariff code
- Quoted exchange rate within 5 percent of the market rate



Finance and control

- Asset (CapEx) or running cost (OpEx)
- Above EUR 250,000: Department Head signs
- Above EUR 1,000,000: CFO and Management Board

// 01 THE COMPANY

Four people, four ways in

Buyer

Pastes a purchase request and reads the audit memo.

Policy owner

Uploads the policy documents the answers come from.

Anyone

Asks what the policy says about a topic, in plain English.

Developer or IT

Watches the system status and the logs.

// SUMMARY

The case in four lines

1 **Aldermoor buys parts from around the world.**
Many suppliers, many currencies, strict internal rules.

2 **Every purchase passes four checks before payment.**
Risk, tax, control and the CFO.

3 **Each check has its own question and its own rules.**
The answers sit in documents and in people's heads.

4 **Why is this hard to do well?**
That is the next question.

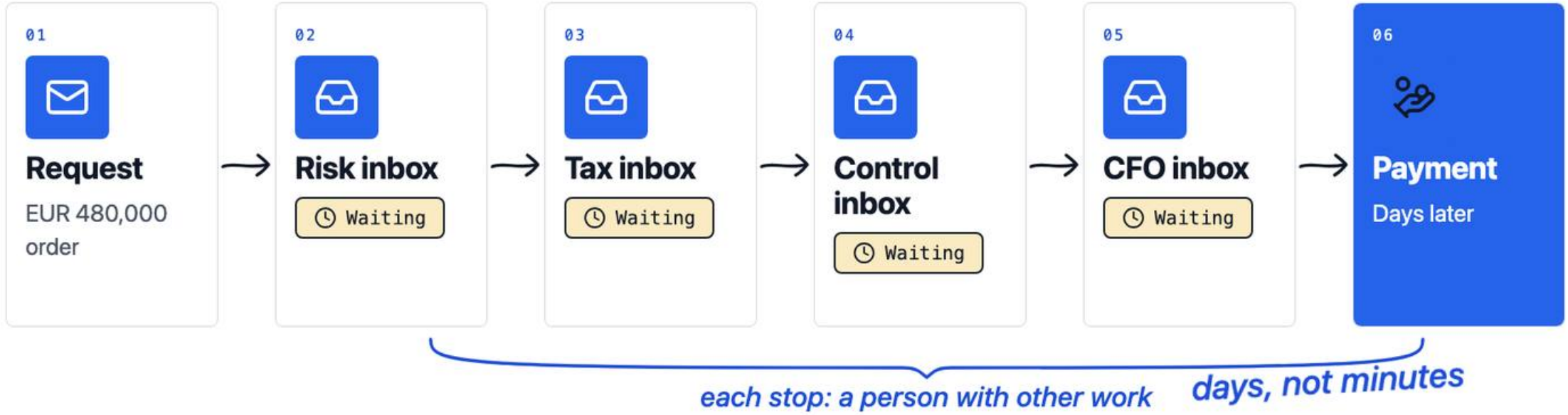
A bigger problem than it looks

Four pains, what they cost, and why the obvious fixes fall short



// 02 THE PROBLEM

One order, four inboxes



// 02 THE PROBLEM

Four pains, one process

It is slow

Each department checks in turn and the request waits in someone's inbox.

The rules are hidden

Payment terms, approval limits and tax rules live in long PDFs.

People differ

Two buyers can get two different answers to the same question.

There is no record

An auditor asks why it was approved and nobody remembers.

// 02 THE PROBLEM

The answer is somewhere in a PDF

"What are our payment terms for servo motor suppliers?"

A buyer asks a simple question. Which document holds the answer, and on which page?

// FOUR DOCUMENTS, FOUR KINDS OF ANSWER

Procurement policy
Payment terms: Net-45 for suppliers with a credit score of 75 or more

EU tariff schedule
Customs duty: 2.2 percent on PLC controllers from Japan

Accounting policy (IFRS)
Depreciation: 10 years, straight line, for filling line equipment

Annual report FY2024
Total revenue: About EUR 2.84 billion

// 02 THE PROBLEM

Two buyers, two answers

// TODAY (ILLUSTRATIVE)

BUYER A

Payment terms for servo motor suppliers? Net-30, I think.

BUYER B

It is Net-45 for suppliers with a credit score of 75 or more.

each answer depends on who read which document, and when

One rule, one answer

// WHAT THE BUSINESS NEEDS

The same input gives the same answer

Whoever asks, whenever they ask.

The answer names its source

So that anyone can verify it.

The decision is written down

So that it can be explained later.

// 02 THE PROBLEM

What a miss can cost

Paying a sanctioned supplier

Legal and regulatory exposure for the company and its managers.

Treating an asset as a running cost

A wrong balance sheet and a wrong depreciation charge.

Accepting an off-market exchange rate

Paying more than the goods are worth.

Having no trail

Not being able to explain a decision when an auditor asks.

Illustrative consequences, not statistics. The aim is to see why speed alone is not enough.

// 02 THE PROBLEM

Why the obvious fixes fall short

OPTION	WHAT IT FIXES	WHAT IT LEAVES OPEN
Reminder emails	A little speed	Rules are still hidden; answers still differ.
A shared folder of PDFs	Rules become findable	Someone must still read and apply them.
A general chatbot	Fast, fluent answers	It has never seen Aldermoor's documents and cannot check a sanctions list.
Fixed rules in code only	Consistency	It cannot read a PDF or judge whether a supplier looks risky.

// 02 THE PROBLEM

A useful mental model

An AI model alone is a brilliant new hire on day one: smart, but it has never seen your files and it cannot pick up the phone.



It has never seen your files

Give it the documents, and let it find the right page. This idea is called retrieval.



It cannot pick up the phone

Give it tools: small functions that look something up or calculate. This idea is called an agent.

// SUMMARY

A good solution must

1 **Be fast.**
Checks run in minutes, not days.

2 **Use Aldermoor's own documents.**
Answers come from the company's rules, not from general knowledge.

3 **Be consistent.**
The same input leads to the same rule and the same answer.

4 **Leave a written record.**
Every step and every decision can be explained later.

5 **Keep people in charge.**
The system prepares; people decide.

// THE APPLICATION

What we are going to build

Two helpers in one application, and how each one behaves



// 03 THE SOLUTION

Two helpers, one application



The document helper

Upload the company's policy PDFs once. Anyone can then ask a question in plain English and get an answer taken from those documents.

Answers questions



The audit helper

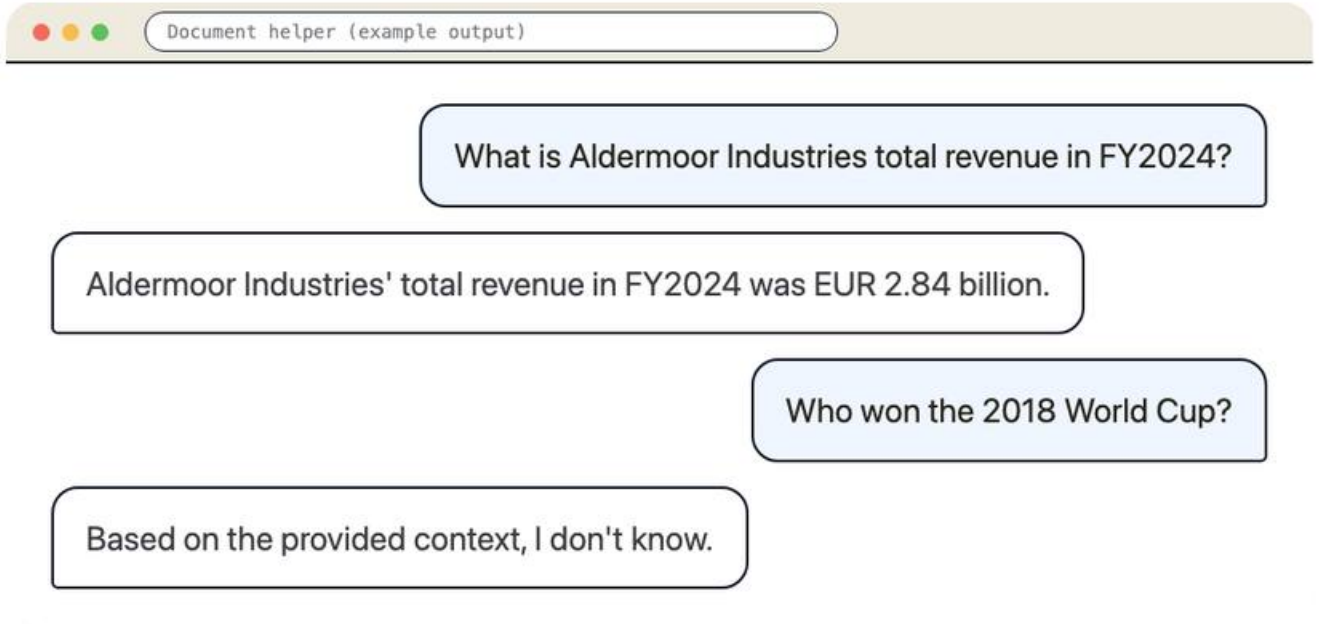
Paste a purchase request. Three specialists check it from different angles and a final step writes one memo with a decision.

Checks a purchase

> The system prepares the answer quickly and consistently, and writes it down. People still make the final call.

// 03 THE SOLUTION

The document helper in use



Answers from the documents

The text comes from the uploaded PDFs, not from general knowledge.

Says it does not know

A question outside the documents gets an honest answer.

Three ways to search

Normal, multi-query and contextual compression.

Example output from a test run, 9 October 2026.

// 03 THE SOLUTION

How the audit works

// THREE SPECIALISTS, EACH WITH THEIR OWN TOOLS

Purchase request

Pasted by the buyer

Risk and compliance

- Sanctions list check
- Vendor credit score

Tax and treasury

- Tax and duty calculator
- Exchange-rate check (live data)

Finance and control

- Asset or running-cost classifier

CFO memo

Reads the three reports and writes one decision

// 03 THE SOLUTION

Decisions follow written rules

The outcome depends on key words in the specialists' reports, not on mood.

Rejected

- Any report contains RED ALERT
- For example, a sanctioned supplier



Conditional hold

- FX ALERT: the quoted rate is more than 5 percent from the market
- or HOLD FOR TREASURY AUDIT



Approved to pay

- All three reports are clear
- The memo confirms the cost centre and ledger account



// 03 THE SOLUTION

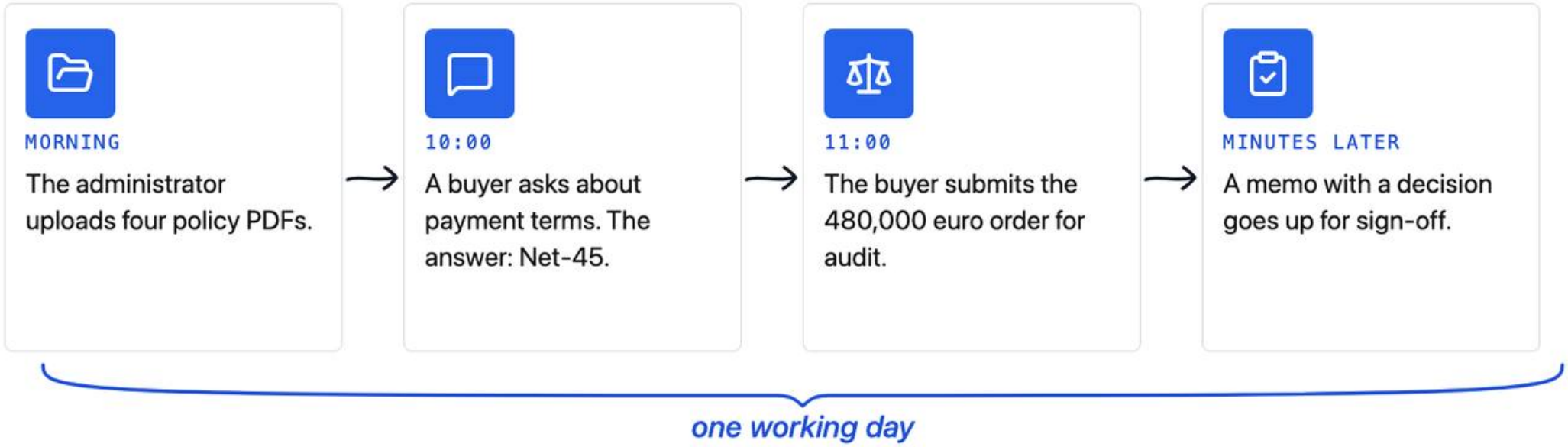
Three audits we will run

SCENARIO	REQUEST	OUTCOME OBSERVED
Approved	20 spare servo motors, EUR 38,000, a well-known supplier, exchange rate within 5 percent of the market	Approved to pay
Hold	200 PLC controllers, EUR 480,000, the fictional supplier, quoted rate 163.5 yen to the euro	Conditional hold: FX alert, high credit risk, amount above EUR 250,000
Rejected	The same order from a sanctioned supplier	Rejected: RED ALERT on sanctions lists

Observed in test runs on 9 and 10 October 2026. Wording varies between runs; the layout and key words do not. Exchange rates move daily.

// 03 THE SOLUTION

A day in the life of a buyer



// 03 THE SOLUTION

What it does not do

LIMIT	WHY IT MATTERS
Four of the five checks use the AI model's general knowledge	A real company connects real sanctions, credit and tax sources. Only the exchange-rate check uses live data.
The audit does not read the uploaded documents	Document questions and the audit are separate features.
AI wording changes between runs	A person reviews important decisions.
There is no sign-in	Anyone with the address can use it. Use only the fictional documents.
Upload history is kept in memory	It resets whenever the server restarts.
Only text PDFs work	A scanned image has no text to read until text recognition is applied.

// 03 THE SOLUTION

How it fits together

// HOW THE PARTS CONNECT

The screen
A web page with four tabs

↓

The server
Receives requests and returns answers

↓

Document helper
Search, then answer

Audit helper
Specialists and memo

↓

The AI model and the stored documents
Gemini, with Chroma on disk when running locally

// BUILT WITH

- Python
- FastAPI
- React
- Docker
- Google Cloud
- GitHub

Plus Gemini as the AI model and LangChain as the toolkit. Each piece is explained when we reach it.

// SUMMARY

What we will build, in four lines

1 **Two helpers in one application.**
One answers questions from documents; one audits a purchase.

2 **Decisions follow written rules.**
Rejected, conditional hold or approved to pay.

3 **People stay in charge.**
The system prepares; people decide.

4 **We know its limits.**
Simulated checks, no sign-in, and wording that varies.

// ROLE VIEW: BUILDING IT

AI ENGINEER

From problem to a working system

How an AI engineer breaks the work down, builds it, and checks it



// 04 AI ENGINEER

What an AI engineer owns



The AI engineer

Designs, builds, tests and ships the AI parts of a system, and proves that they work.

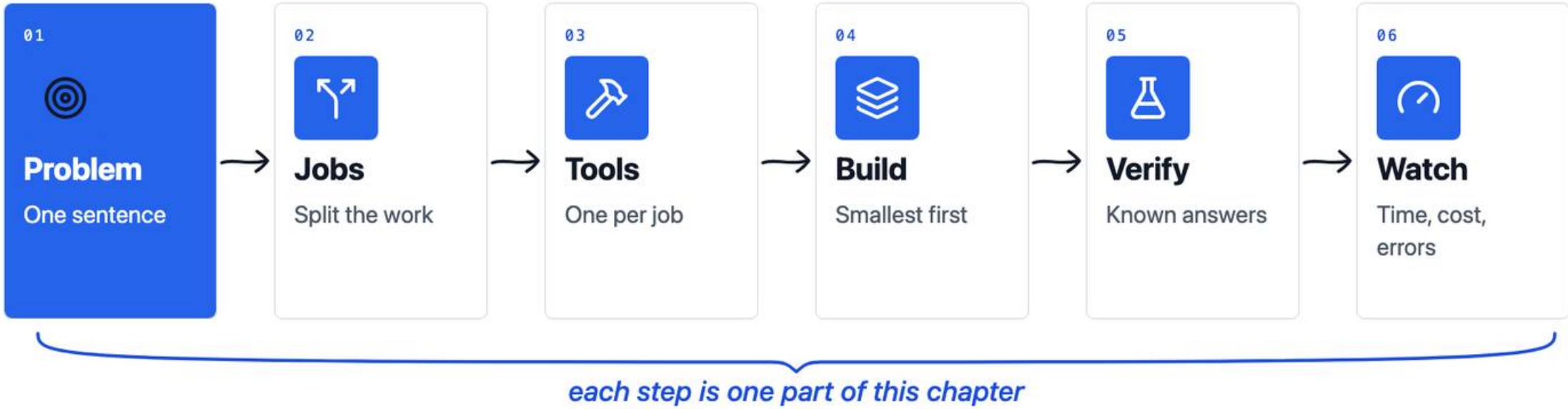
Owns
The model calls, prompts, tools and retrieval: the AI parts of the system.

Asks
Is it correct? Is it safe? Can we test it? What will it cost?

Delivers
A working service that has been tested, logged and packaged.

Measures
Accuracy, speed, cost and how often a person has to step in.

The route in six steps



Start with the problem, not the model

Make the checks faster, make them consistent, and leave a written record, without taking the decision away from people.

Faster | Consistent | Recorded | People decide

> Every part of the project has to serve this sentence. If it does not, it is not built.

Break it into six jobs

`rag/`

Find the right rule
Search the documents by meaning

`agent/`

Check from several angles
Specialists with their own tools

`agent/prompts.py`

Decide by written rules
Key words that lead to one of three outcomes

`api/ + frontend/`

Let people use it
A web service and a screen

`logger/ + tests/`

Make it trustworthy
Logs, clear errors, limits and tests

`Dockerfile + deploy.yml`

Reach other people
Package it and put it online

> The tag on each job is the folder or file in the project where it will live. We build them one at a time.

One giant prompt



Hard to test
A failure could be anywhere inside it.

Costly, with no visible source
Every question carries everything.

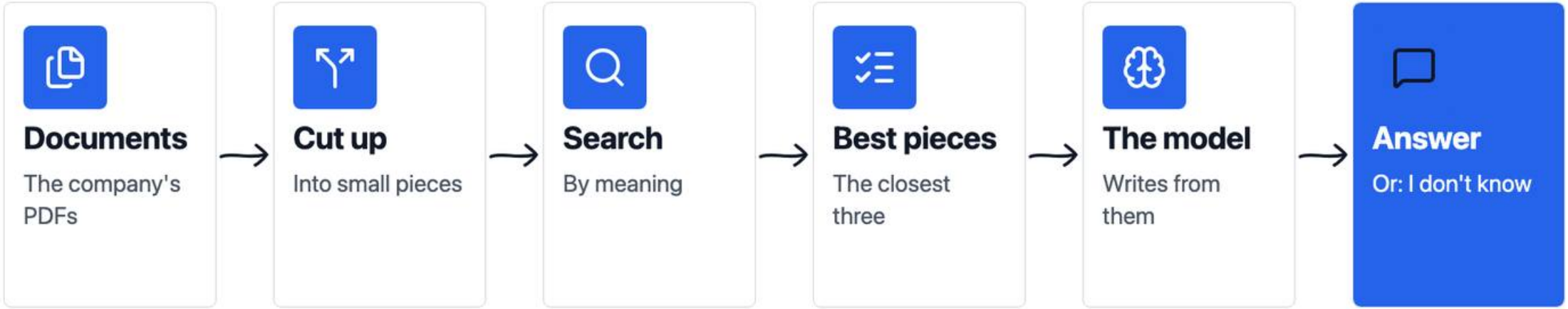
One tool per job

- Search
- Risk
- Tax
- Control
- Decide
- Log

Each part is tested alone
A problem has an address.

Replaceable, with sources visible
Swap a search method or a model.

Search first, then answer



Paste every document in: slow, costly, no source

Search first: fast, focused, the source stays visible

> This idea is called retrieval-augmented generation. We build it step by step.

Exact rules in code, judgement for the rest

// EXACT AND CHECKABLE: ORDINARY CODE

exchange-rate check

```
# agent/tools.py  (the exchange-rate check)
variance = abs(rate_used - market_rate) / market_rate

if variance > 0.05:
    return "FX ALERT: variance exceeds 5% limit"
return "FX SUCCESS: within the 5% band"
```

E-mail addresses removed before the model

Card numbers masked

// NEEDS READING AND REASONING: THE MODEL

Does this supplier look risky?

Is this item a lasting asset or a running cost?

the skill is knowing which is which

Build the smallest thing first

at every step, it still runs →

STEP 1

Notebook sketch

Does the idea work at all?

STEP 2

Function

Clear input, clear output.

STEP 3

Named file

One file, one job.

STEP 4

Settings outside

Keys and names in .env.

STEP 5

Safety layer

Logs, errors, tests.

Check against known answers

// PARTS WE CONTROL: AUTOMATED TESTS

terminal

```
$ pytest backend/tests
..................                  [100%]
17 passed in 7.47s
```

No key and no internet needed. The model is replaced by a stand-in.

// PARTS THE MODEL WRITES: KNOWN QUESTIONS

QUESTION	EXPECTED	OBSERVED
Total revenue in FY2024	EUR 2.84 billion	EUR 2.84 billion
Payment terms, servo motor suppliers	Net-45	Net-45
Customs duty, PLC controllers from Japan	2.2 percent	2.2 percent
Depreciation, filling line equipment	10 years	10 years, straight line
Who won the 2018 World Cup?	I don't know	I don't know

Observed in a test run, 9 October 2026.

Three things to watch



Time and cost

An audit took roughly 70 to 150 seconds in testing and made many model calls. Cap the calls.



Confident mistakes

A model states wrong things as fluently as right ones. A person reads the result.



Rules hidden in text

The final decision rule lives inside a prompt. A production system checks the key words in code.

// SUMMARY

The AI engineer's route in five lines

1 **Start with one sentence of the problem.**
Faster, consistent, recorded, people decide.

2 **Split it into jobs and pick a tool for each.**
Search, specialists, rules, service, safety, delivery.

3 **Build the smallest working thing first.**
Notebook, function, file, settings, safety.

4 **Check against known answers.**
Test what you control without the model.

5 **Watch time, cost and confident mistakes.**
Keep a person in the loop.

// ROLE VIEW: REAL USE

FORWARD-DEPLOYED ENGINEER

From working system to real use

How a forward-deployed engineer understands the people, connects the data and hands it over



What a forward-deployed engineer owns



The forward-deployed engineer

Works inside the customer's organisation to turn a working system into daily use.

Starts from

The people, the process and the constraints at the customer.

Asks

What do you do today? What data exists? What will you trust?

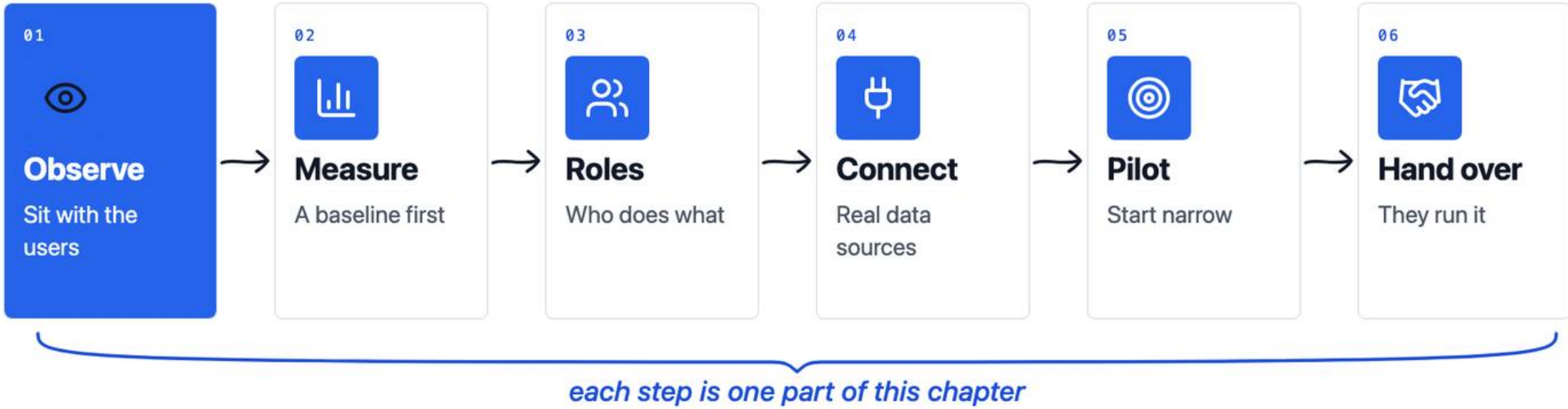
Delivers

The system in daily use, plus a handover the team can run.

Measures

Time saved, adoption, and trust, against a baseline.

The route in six steps



Start with the people

Sit with a buyer, a tax specialist and a controller. Ask them to show you what they did yesterday.

The real path

Not the one on the process chart.

Where it waits

Which step holds the order longest?

What comes back

Who sends it back, and why?

Repeated questions

The ones a system could answer.

Measure the baseline first

Days from request to payment

today: _____

after: _____

Hand-offs between people

today: _____

after: _____

Requests sent back to the buyer

today: _____

after: _____

Decisions nobody can explain later

today: _____

after: _____

> Without a baseline, "it feels faster" is the only evidence. With one, the result is a number. The blanks belong to the customer.

// 05 FORWARD-DEPLOYED ENGINEER

The system prepares

Looks up the rules
In the policy documents.

Runs the checks
Risk, tax and control.

Writes the memo
One decision with reasons.

Records every step
So it can be explained.

People decide

Approve the payment
A person signs off.

Resolve the holds
Treasury and procurement act.

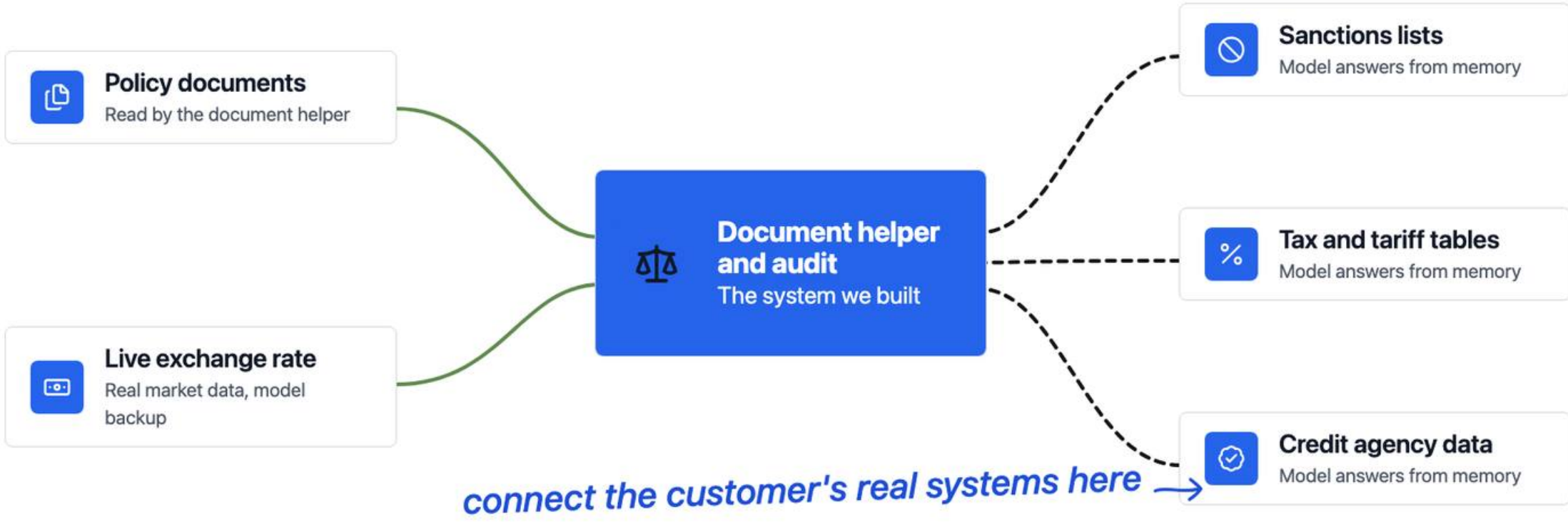
Handle the exceptions
What the rules did not foresee.

Own the policy
Rules change; people change them.

Where the real information lives

READ FOR REAL

SIMULATED BY THE MODEL TODAY



Two answers that disagree

// THE AUDIT'S TAX CHECK

0 %

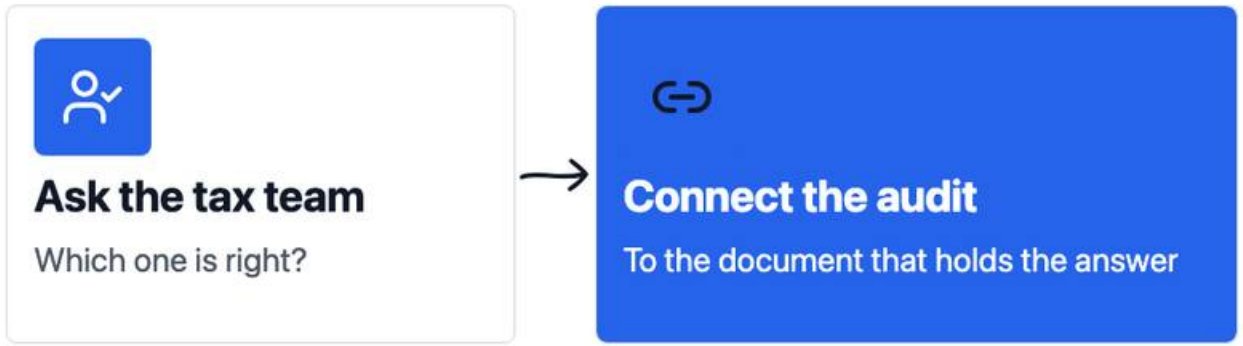
Customs duty on Japanese PLC controllers, from the AI model's general knowledge.

≠

// THE COMPANY'S TARIFF DOCUMENT

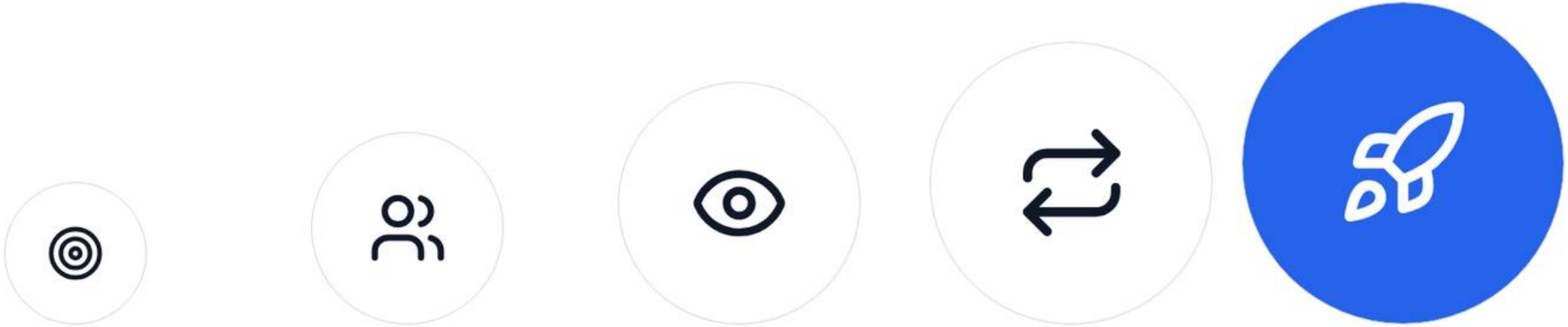
2.2 %

The same duty, found by the document helper in the tariff schedule.



Observed in a test run, 9 October 2026.

Start narrow, then widen



One slice
One category, one team.

A few real users
Early, even if rough.

Watch
How they actually use it.

Fix and measure
Against the baseline.

Widen
Only after it works.

widen only with evidence, never before

Work inside the customer's rules



Privacy

E-mail addresses are redacted and card numbers masked before text reaches the model. Free-tier model content may be used by the provider, so a demonstration uses fictional documents only.



Access

The demonstration has no sign-in. A real deployment needs sign-in, roles and an approval trail that security will accept.



Cost

Some cloud parts bill by the hour even when idle. Agree a budget, set an alert, and know how to switch everything off.

Make it trusted, then hand it over

// EVERY AUDIT LEAVES A RECORD

logs/…log (illustrative format)

```
{"event": "Tool call started",
 "tool": "validate_fx_hedge",
 "args": {"currency_pair": "EUR_JPY", "rate_used": 177.0}}
{"event": "Tool call finished",
 "tool": "validate_fx_hedge", "duration_ms": 105}
```

> If an auditor asks why something was approved, someone can answer.

// BEFORE LEAVING, THE CUSTOMER'S TEAM CAN

- ✓ Run the system
- ✓ Change a prompt
- ✓ Read the logs
- ✓ Switch everything off

// 05 FORWARD-DEPLOYED ENGINEER

Two lenses, one project

	AI ENGINEER	FORWARD-DEPLOYED ENGINEER
Starts from	A defined problem and some data	People, process and constraints
Main question	Does it work, and can we prove it?	Will people use it and trust it?
Spends time on	Prompts, tools, retrieval, tests	Observation, data access, rollout, training
Delivers	A tested, working system	The system in daily use, plus a handover
Success looks like	Correct, fast and affordable	A measured gain against the baseline

// SUMMARY

Real-world use in five lines

1 **Start with the people and measure a baseline.**
Observe the real path before choosing technology.

2 **The system prepares; people decide.**
Say it early and mean it.

3 **Find where the real information lives.**
Replace simulated checks with real sources.

4 **Start narrow and widen with evidence.**
One slice, a few users, repeated measurement.

5 **Respect privacy, access and cost; keep a record.**
The customer's rules come first, and the team can run it without you.